

Due: Nov. 4 2025, 20:00pm

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作业一：了解上市公司 **Know public company.**

上网寻找原始资料（财报，SEC文件等）回答有关电子信息产业企业的问题：

Search online for original data (financial reports, SEC filings, etc.) to answer questions about companies in the electronic information industry:

挑选 3 家你喜欢的美股上市的电子信息公司（建议是电子硬件或电路公司如NVDA, AVGO, TXN等）,3家在 A 股上市的电子信息公司，他们分别属于电子信息产业分类中的哪一细类？为什么？它的年产值(Revenue)有多少，年利润(Profit)有多少，整个公司值多少钱，为什么？最近一个财政季度(quarter)它们各自的产值和利润又是多少？收集研究公司资料，说明数据来源，要求图文并茂。

Pick three electronics companies you like that are listed on the US stock market (we recommend electronic hardware or circuit companies such as NVDA, AVGO, and TXN) and three electronics companies listed on the A-share market. Which sub-category of the electronics industry do they belong to? Why? What is their annual revenue and profit? What is the overall company value? Why? What were their respective output and profits in the most recent fiscal quarter? Collect and research company data, explain the data sources, and provide both pictures and text.

作业二：了解 Shell

基于树莓派，完成 Shell（建议使用 bash 或 csh）编程要求如下，要求精确读懂题意并编程。
需提交运行视频，并在作业中提交实验报告（截图）。

Based on the Raspberry Pi, complete the following Shell programming requirements (bash or csh recommended). Precisely understand the assignment and implement the programming accordingly.
Submit a video demonstrating the program in operation, along with a lab report (screenshots) in the assignment submission.

Write a shell script (using vi or other editors you like) that displays the following menu and prompts for one-character input to invoke a menu option, as shown.

- a. List all files in the present working directory
- b. Display today's date and time
- c. Display whether a file is a "simple" file or a "directory"
- d. Create a backup for a file using "tar" command
- e. Compress the tar file
- f. Start an ftp session
- g. Start your LED control program**
- x. Exit

For options (c) and (d), prompts the user for file names before invoking a shell command/program. For option (e), prompts the user for a domain name (or IP address) before initiating an ftp session. The program should allow the user to try any option any number of times and should quit only when the users give option x as input.

注：以上功能为基本功能，建议拓展更多菜单内容，可作为加分项；

Note: The above functions are basic functions. It is recommended to expand more menu content as a bonus item;

作业三：了解Makefile

基于树莓派，完成 Shell（建议使用 bash 或 csh）编程要求如下，要求精确读懂题意并编程。需提交运行视频，并在作业中提交实验报告（截图）。

Using a Raspberry Pi, complete Shell programming (bash or csh is recommended). The following requirements are required. You must accurately understand the problem and program the code. A running video and a lab report (with screenshots) must be submitted as part of the assignment.

Write a Makefile, include these functions:

1. **Make sort_file:** list the file/directories with respect to the file size
2. **Make find_log:** find the log file in your home directory (you may create some dummies)
3. **Make delete_lock_file:** find the lock files in your home directory and then delete them.

作业要自己完成，本课程对抄袭零容忍，2021、2022、2023、2024学年已出现作业抄袭导致零分处理的情况。

Assignments must be completed by yourself. This course has zero tolerance for plagiarism. In the 2021, 2022, 2023, and 2024 academic years, there have been cases where plagiarism in assignments resulted in zero marks.